

Polygenic Risk Score Computation with PRSice

Emma S. Lockett, PhD

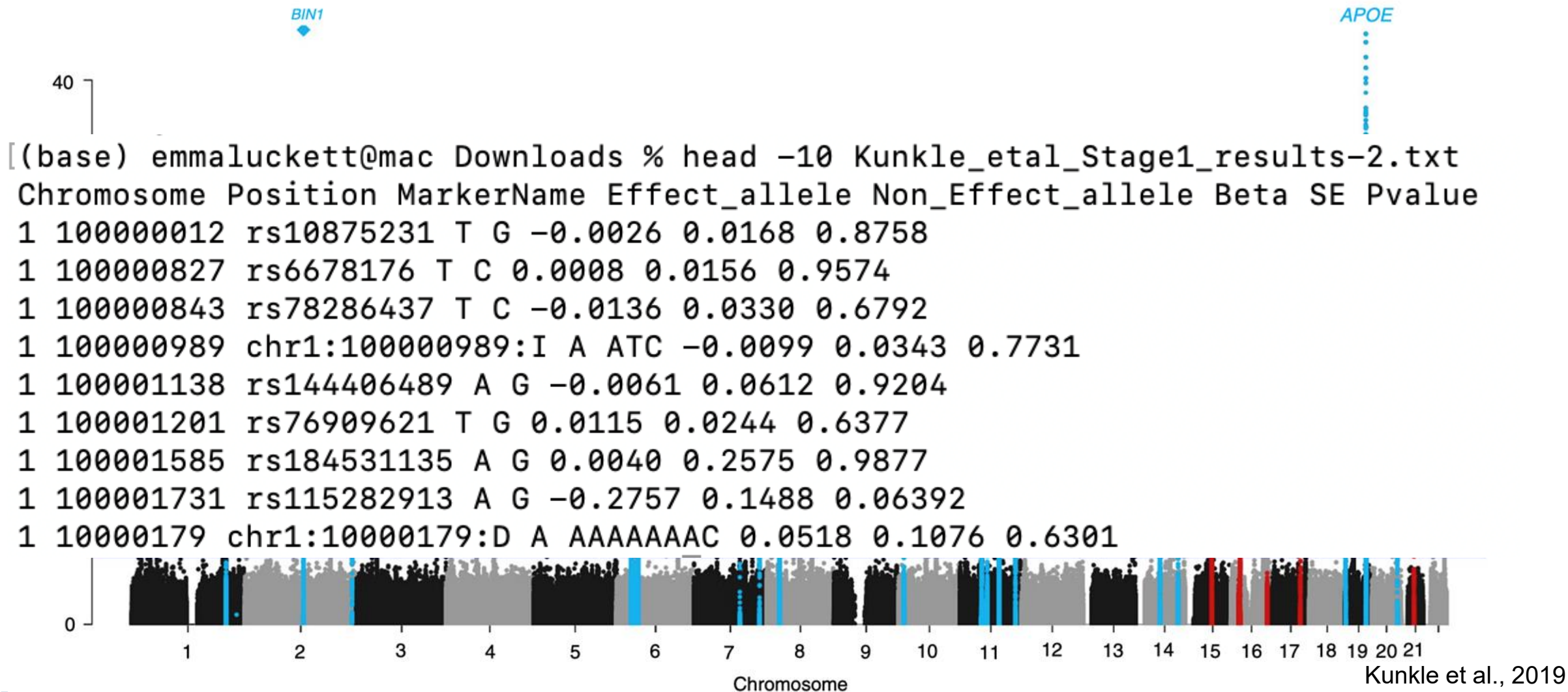
Amsterdam UMC

25.07.2025

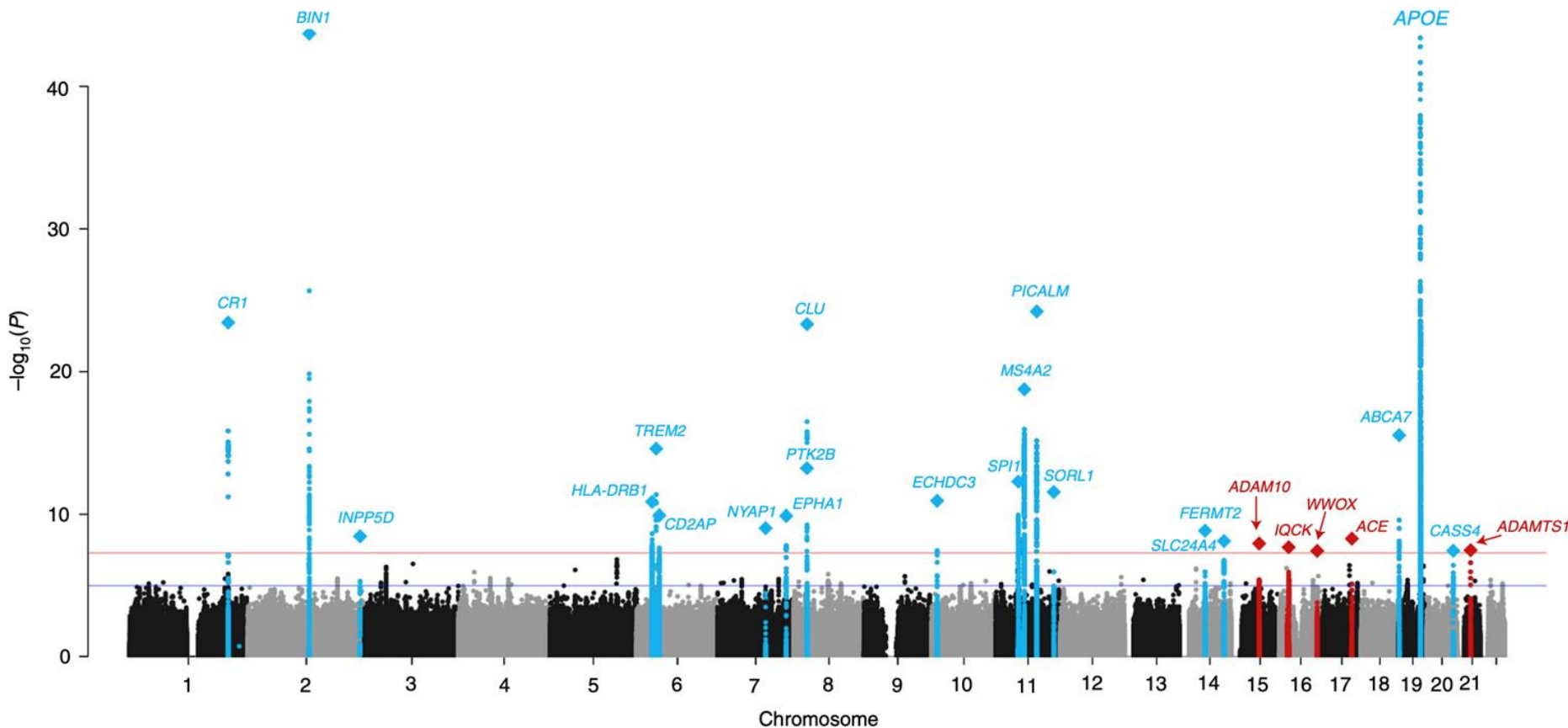


1. Recap
2. What is a polygenic risk score and the limitations
3. Why compute a polygenic risk score
4. Summary of tools
5. PRSice workflow, summary statistics, and input data
6. PRSice output
7. ADNI example

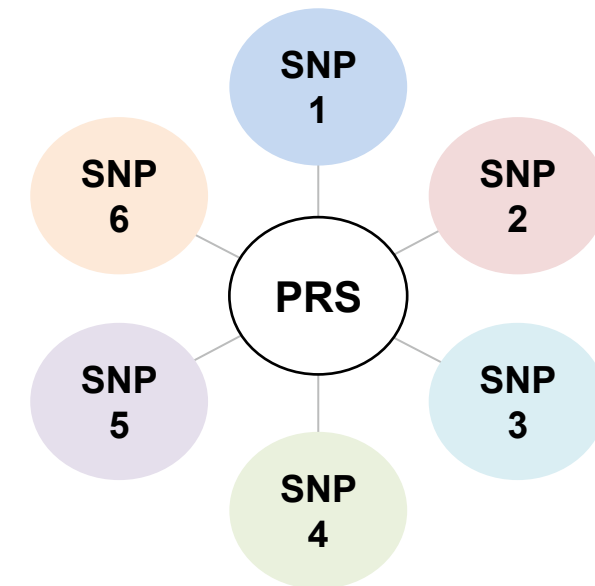
- So far, you've learnt:
 - How to design a genetics experiment
 - How to obtain (genetic) data and perform quality control
 - SNP analysis
 - How to conduct a GWAS



What is a Polygenic Risk Score (PRS)?



Kunkle et al., 2019



Formula:

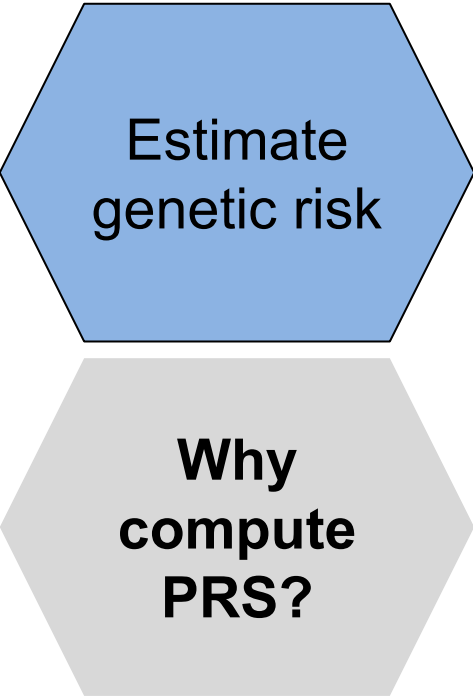
$$\text{PRS} = \sum (\text{effect size} \times \text{genotype})$$

- Modest predictive power
- Limited cross-ancestry generalisation
- Ignores environmental and epigenetic effects



**Why
compute
PRS?**

Why use PRS?

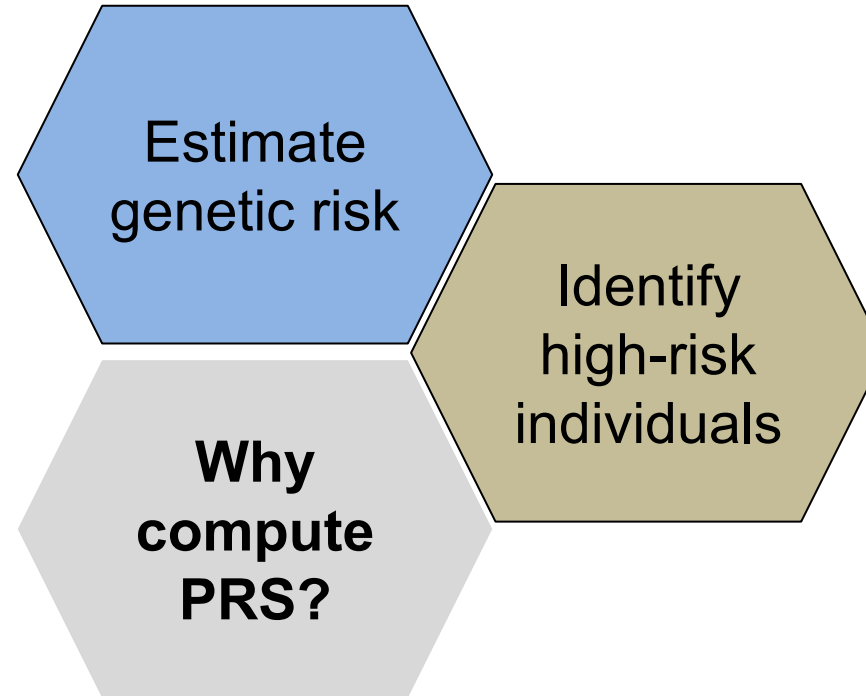


Estimate
genetic risk

**Why
compute
PRS?**

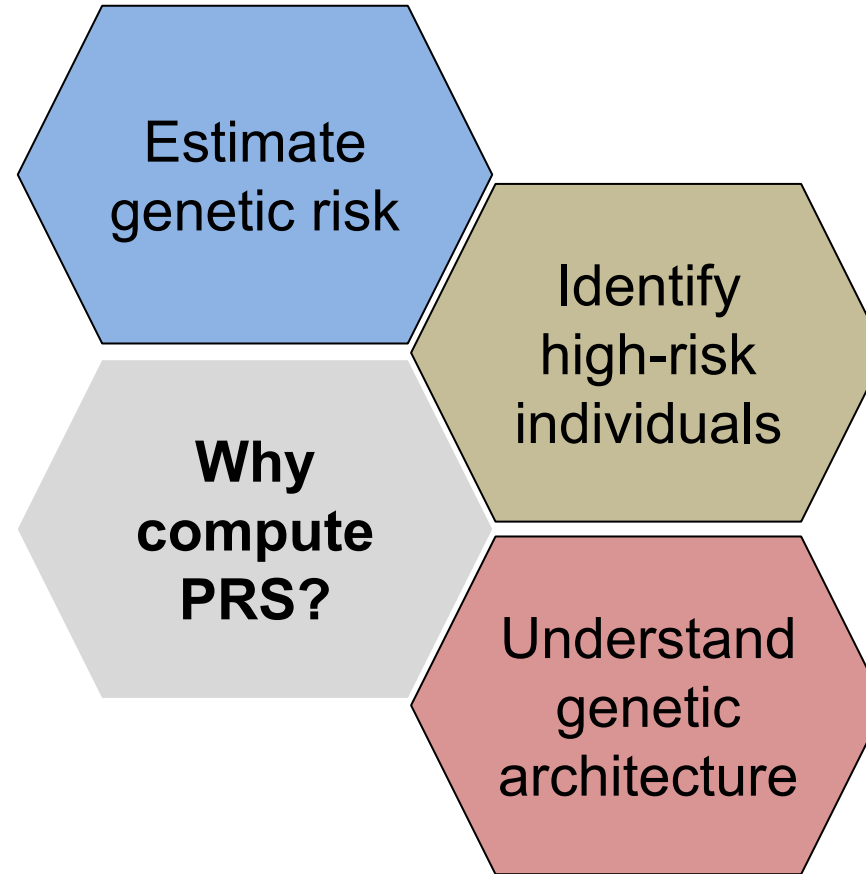
- PRS aggregates the effects of thousands of variants (SNPs) into a single number that reflects an individual's risk for a trait or disease

Why use PRS?



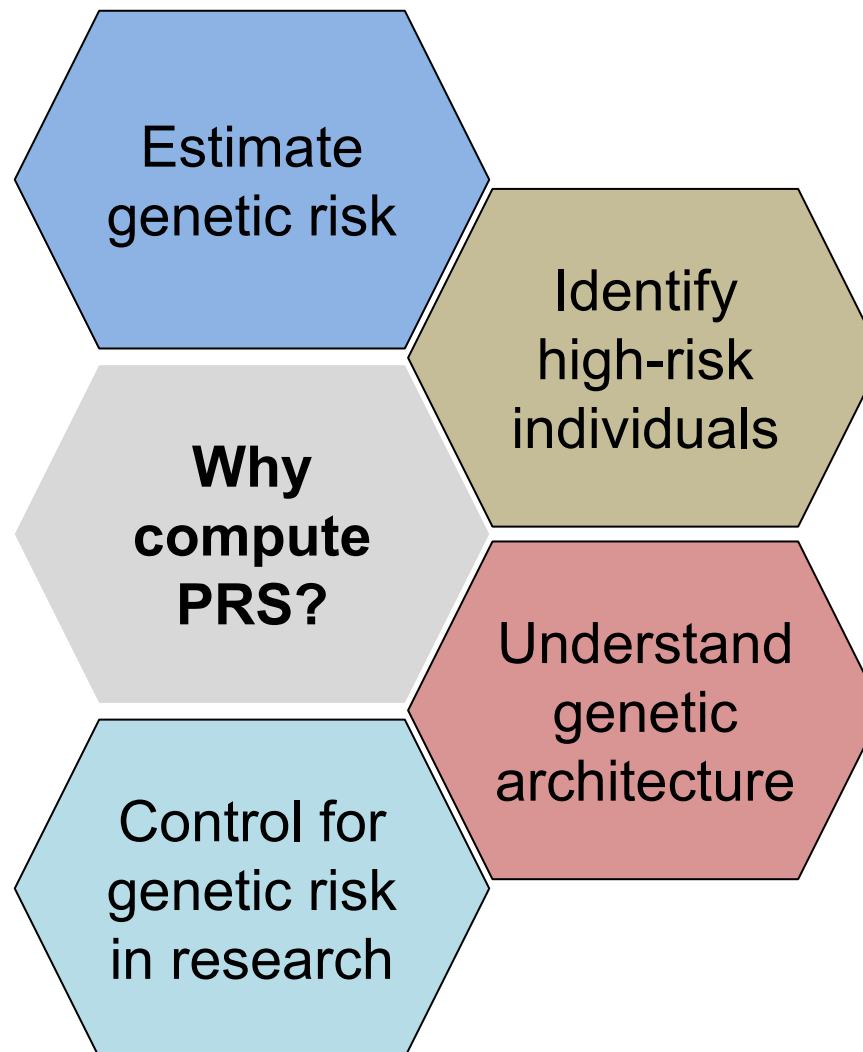
- People with high PRS may have substantially higher risk compared to the population average
- This can help with:
 - Early screening
 - Preventive interventions
 - Personalised medicine

Why use PRS?



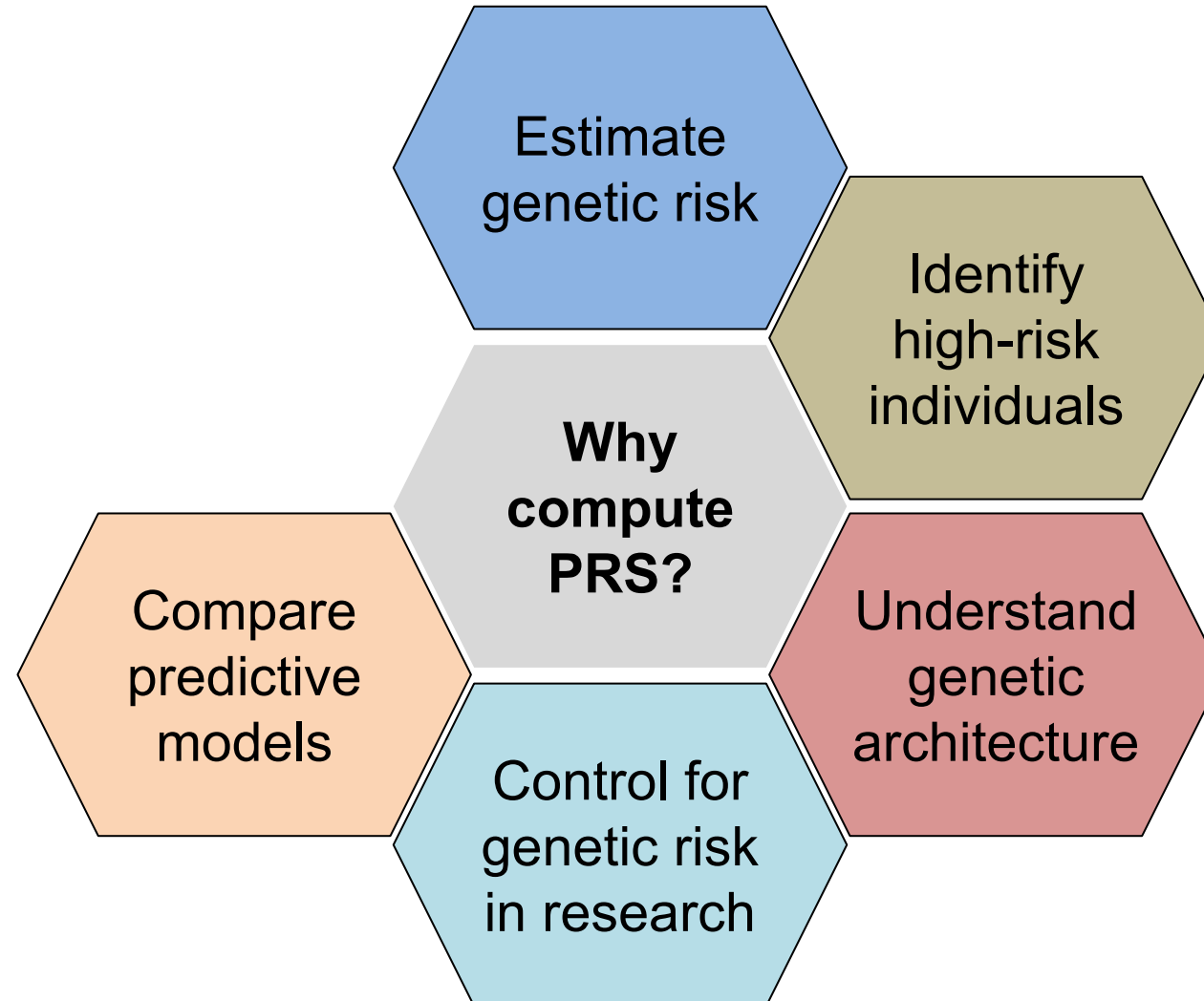
- PRS can help researchers understand how much of a trait is explained by common genetic variation and how different sets of variants contribute to it

Why use PRS?



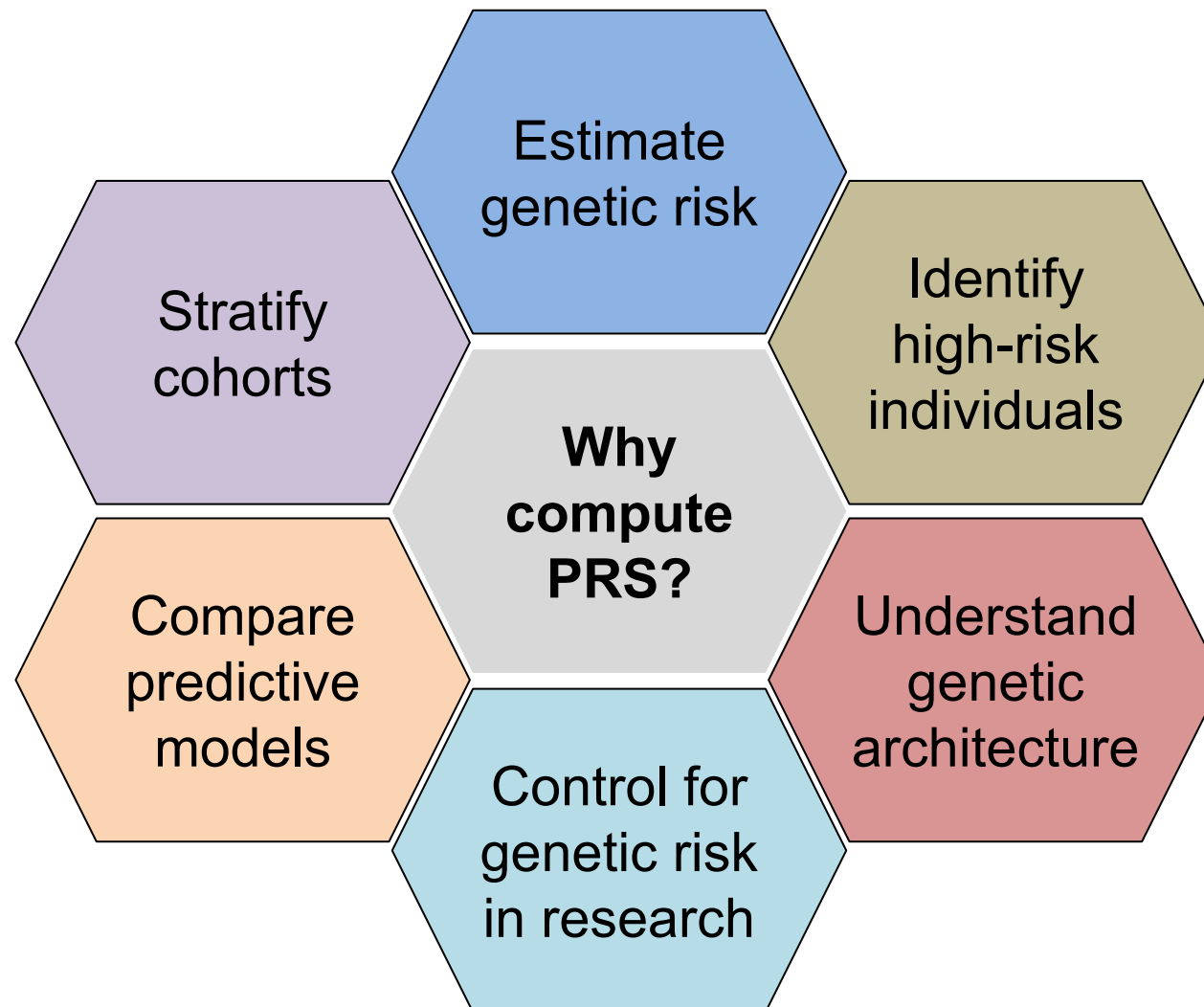
- In studies of e.g. brain imaging, cognition, or biomarkers, PRS can be used as a covariate to control for genetic risk or to explore how genetics influence these traits

Why use PRS?



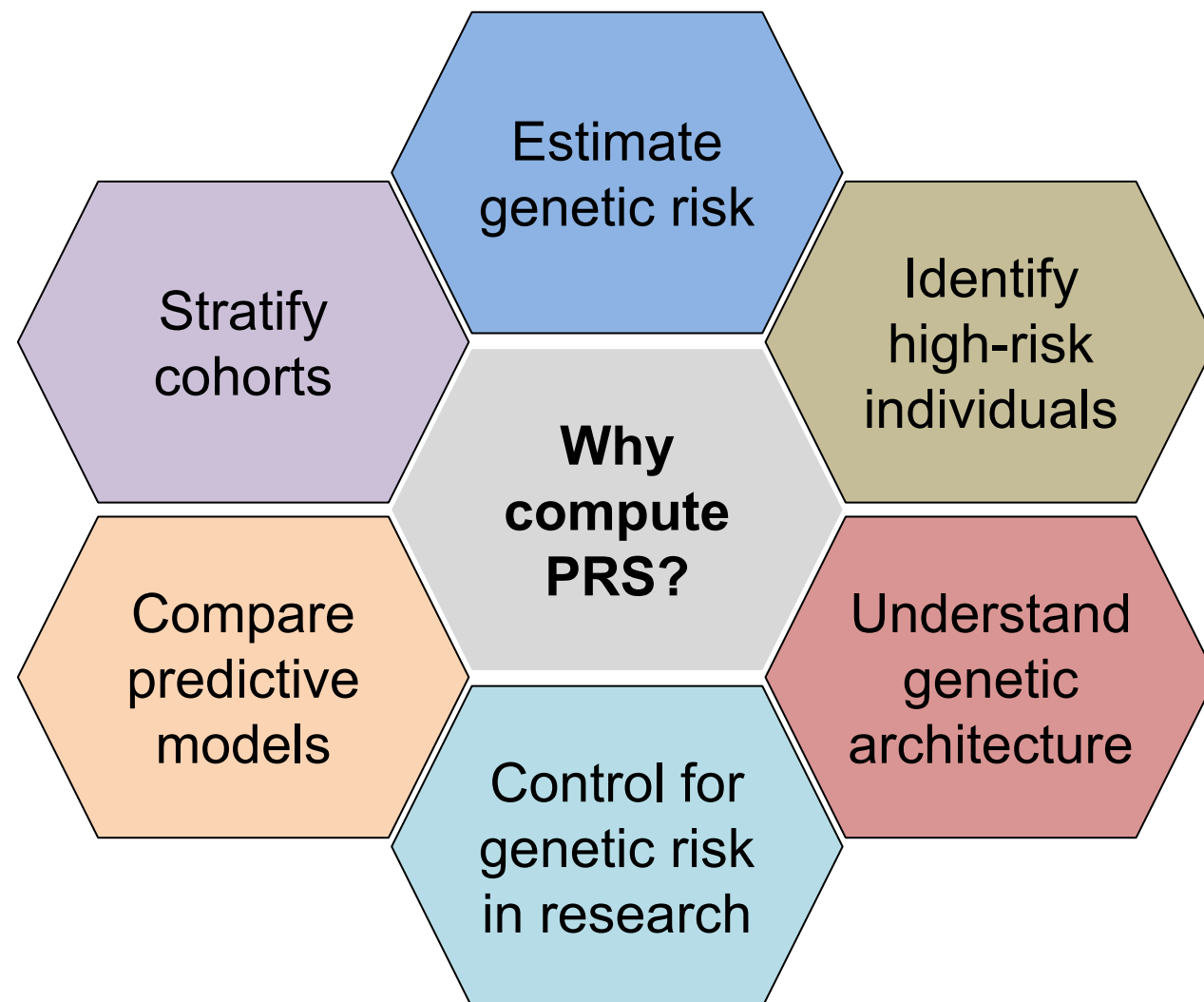
- PRS allows comparison of different genetic models (e.g., including or excluding APOE) to test how well genetic data predict traits

Why use PRS?



- PRS can be used to stratify participants into different risk groups, which is especially useful in:
 - Clinical trials
 - Longitudinal studies
 - Prevention research

Why use PRS?



- **Important considerations:**
- Ancestry-matched base and target data
- Include population stratification covariates (genetic PCs)
- Validate in independent datasets
- Use multiple thresholds for SNP inclusion (pT)

PRS computation tools

Method	Models LD?	Shrinkage?	Threshold tuning?	Speed	Language	Best use case
LDpred2	✓	✓	✗ (auto)	Moderate	R	Accurate modelling of LD, multiple PRS models
PLINK	✗	✗	✓	✓ ✓	C++	Fast, large datasets, simple baseline
PRSice	✗	✗	✓ (auto tested)	✓ ✓	R + C++	Easy, fast PRS screening across thresholds
PRS-CS	✓	✓	✗	Moderate	Python	No threshold tuning, good polygenic modelling
SBayesR	✓	✓	✗	Slow–Moderate	Python/C++	Advanced modelling, large/complex GWAS

Method	Models LD?	
LDpred2		- GWAS tests each SNP <i>individually</i> for association with a trait (e.g., Alzheimer's risk, a beta coefficient and a p-value) Assumption: <i>each SNP is tested independently</i> But this independence assumption is not true in reality
PLINK		So LDpred2 aims to recover true SNP effects using a <i>Bayesian model</i> that adjusts GWAS effect sizes for linkage disequilibrium (LD)
PRSice		- Start with prior beliefs about SNP effect sizes: - Example: Most SNPs have zero effect, a few have small/medium/large effects. - In LDpred2: prior = mixture of Gaussians - Use the observed GWAS summary statistics (betas and p-values) and an external LD matrix (correlations between SNPs from a reference panel) - Update the effect size estimates based on: - How strong the GWAS signal is - How correlated that SNP is with others - How likely it is, under the prior, that the SNP has a real effect
PRS-CS		
SBayesR		This process "shrinks" noisy effect sizes towards their true value

Method	Models LD?	Shrinkage?
LDpred2	✓	✓
PLINK	✗	✗
PRSice	✗	✗
PRS-CS	✓	✓
SBayesR	✓	✓

- GWAS effect sizes are often **inflated**, due to:
 1. Sampling noise (especially in small samples)
 2. Linkage disequilibrium (LD)
 3. Winner's curse (top SNPs look stronger than they are)
- Shrinkage estimates a more conservative value**
- True effects are retained (e.g., SNP1 stays positive) and false positives are shrunk toward zero (e.g., SNP2 is penalised)
- PRS-CS uses *Bayesian continuous shrinkage* that assumes most SNPs have no or small effect and updates the effect sizes using the GWAS data and the LD structure:
 1. Keep true effect sizes (even if small)
 2. Shrink noisy or false signals toward zero
 3. Do all this without altering p-value thresholds
- Continuous shrinkage process adaptively shrinks the effect sizes based on how strong the GWAS signal is for a particular SNP, and the LD of that SNP with others**

Method	Models LD?	Shrinkage?	Threshold tuning?
LDpred2	✓	✓	✗ (auto)
PLINK	✗	✗	✓
PRSice	✗	✗	✓ (auto tested)
PRS-CS	✓	✓	✗
SBayesR	✓	✓	✗

• *p-value thresholding* implemented in PRSice is the process of selecting SNPs based on their GWAS p-values:

1. Setting up a range of p-value thresholds (e.g., 5e-8 to 1.0)
1. Keeping all SNPs within the different GWAS $p \leq$ threshold
1. Clumping SNPs in LD (to keep independent SNPs) and calculates a PRS for each individual
1. Testing how well different sets of SNPs predict the phenotype using a regression
1. Recording R^2 and p-value and plotting R^2 vs. p-threshold to find the optimal threshold for SNP inclusion
1. Choosing the best-performing threshold (e.g., $p < 0.01$, $p < 0.05$, $p < 0.5$, etc.)

PRS computation tools

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Input data for PRS calculations with PRSice



GWAS summary statistics = base file

The file with GWAS summary statistics



Genotype data = target data

The prefix of the files that contain the genotype data in binary plink format



Optional: Phenotype file

FID – Family ID (usually same as IID if not using family data)

IID – Individual ID

Phenotype – Your target trait or disease status (binary or continuous)



Optional: Covariates file

File containing genetic principal components or other covariates such as age, as necessary



Optional: External dataset for clumping

Within each block of correlated SNPs, the SNP with the lowest p-value in the discovery set is selected

How to obtain GWAS summary statistics: GWAS catalogue example


GWAS Catalog

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GWAS Catalog

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Authors Kunkle BW , Grenier-Boley B, Sims R, Bis JC, Damotte V, Naj AC, Boland A, Vronskaya M, van der Lee SJ... [Show more >](#)

Full Summary Statistics 

Available data:

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Studies with summary statistics 1 

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First author	Study accession	Pub. date	Journal	Title	Reported trait	Trait(s)	Background trait(s)	 Discovery sample number	 Replication sample number	Association count	Summary statistics
Kunkle BW	GCST007511	2019-02-28	Nat Genet	Genetic meta-analysis of diagnosed Alzheimer's...	Alzheimer's disease (late onset)	late-onset Alzheimers disease	-	• 63926 European	• 30511 European	24	FTP Download

Showing 1 to 1 of 1 entries

«
1
»

Index of /pub/databases/gwas/summary_statistics/GCST007001-GCST008000/GCST007511

Name	Last modified	Size	Description
Parent Directory	-	-	-
Kunkle et al 2019 IGAP summary statistics README 0.docx	2019-08-14 00:02	16K	
Kunkle et al Stage1 results.txt	2019-08-14 00:02	543M	

*.fam

FID	IID	PID	MID	Sex	P
1	1	0	0	2	1
2	2	0	0	1	0
3	3	0	0	1	1

*.bed

Contains binary version of the SNP info of the *.ped file. (not in a format readable for humans)

*.bim

Chr	SNP	GD	BPP	Allele 1	Allele 2
1	rs1	0	870000	C	T
1	rs2	0	880000	A	G
1	rs3	0	890000	A	C

Chromosome	Position	MarkerName	Effect_allele	Non_Effect_allele	Beta	SE	Pvalue
1	100000012	rs10875231	T G	-0.0026 0.0168	0.8758		
1	100000827	rs6678176	T C	0.0008 0.0156	0.9574		
1	100000843	rs78286437	T C	-0.0136 0.0330	0.6792		
1	100000989	chr1:100000989:I	A ATC	-0.0099 0.0343	0.7731		
1	100001138	rs144406489	A G	-0.0061 0.0612	0.9204		
1	100001201	rs76909621	T G	0.0115 0.0244	0.6377		
1	100001585	rs184531135	A G	0.0040 0.2575	0.9877		
1	100001731	rs115282913	A G	-0.2757 0.1488	0.06392		
1	10000179	chr1:10000179:D	A AAAAAAC	0.0518 0.1076	0.6301		
1	100002106	rs17120619	C G	0.4699 0.2869	0.1015		
1	100002154	chr1:100002154:D	T TGTTA	0.0114 0.0244	0.6405		
1	100002155	chr1:100002155:D	G GTTAGT	0.0114 0.0244	0.6406		
1	100002490	rs78642210	T C	0.0149 0.0331	0.6523		
1	100002713	rs77140576	T C	0.0061 0.0237	0.7982		
1	100002714	rs113470118	A G	-0.0150 0.0331	0.651		
1	100002882	rs7545818	T G	-0.0015 0.0156	0.9241		
1	100002991	rs75635821	A G	0.0060 0.0237	0.7992		
1	100003204	rs78948828	T G	-0.0150 0.0331	0.6507		
1	100003419	rs114427610	T C	-0.0128 0.0487	0.7924		
1	10000400	rs1237370	A T	-0.0094 0.0210	0.6545		
1	100004203	chr1:100004203:I	G GT	-0.0253 0.0542	0.6412		
1	100004204	chr1:100004204:I	T TTTTGT	-0.0091 0.0198	0.6443		
1	100004210	chr1:100004210:I	T TTTTGT	-0.0128 0.0194	0.5089		
1	100004463	chr1:100004463:D	T TA	-0.0017 0.0168	0.9195		
1	100004465	chr1:100004465:D	A AT	0.0042 0.0178	0.8126		
1	100004726	rs6682190	A G	-0.0111 0.0208	0.5918		
1	100004916	chr1:100004916:D	G GATT	-0.0190 0.0198	0.3389		
1	100005230	rs6697069	A T	-0.0103 0.0235	0.6597		
1	100005477	rs12069019	A G	-0.0111 0.0208	0.5923		
1	100005950	rs150684236	A G	-0.0852 0.0945	0.3673		
1	100006117	rs6686057	A G	0.0193 0.0147	0.1911		
1	100006734	rs55725529	T C	-0.0090 0.0215	0.6748		
1	100007258	rs76698872	T C	0.0026 0.0452	0.9538		
1	100007454	rs12082355	T C	-0.0109 0.0208	0.6006		
1	100007741	rs12067343	A G	0.0109 0.0208	0.6012		
1	100007961	rs35363137	A G	-0.0969 0.0758	0.2015		
1	100008607	rs11166268	A C	-0.0016 0.0156	0.9208		
1	100008708	chr1:100008708:D	T TG	0.0238 0.0224	0.2866		
1	100008737	rs188491891	C G	-0.0377 0.1199	0.753		
1	100008943	rs149181078	T G	0.0224 0.0614	0.7156		
1	100008987	rs11166269	A C	0.0012 0.0156	0.9364		
1	100008993	rs12039860	C G	0.0944 0.2601	0.7167		
1	100009669	rs6698430	T C	-0.0012 0.0156	0.9393		
1	100010065	rs112013596	T C	-0.0155 0.0330	0.6397		
1	100010434	rs12130109	A G	0.0054 0.0237	0.8194		
1	100010753	chr1:100010753:D	T TAACGAC	0.4283 0.2316	0.04703		

Basic Tutorial for Polygenic Risk Score Analyses

- Home
- 1. QC of Base Data
- 2. QC of Target Data
- 3. Calculating and analysing PRS PLINK
- PRSice-2
 - Obtaining PRSice-2
 - Required Data
 - Running PRS analysis
- LDpred-2
- lassosum
- 4. Visualizing PRS Results

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[Docs](#) » PRSice-2 [Edit on GitHub](#)

Background

PRSice-2 is one of the dedicated PRS programs which automates many of the steps from the previous page that used a sequence of PLINK functions (plus some QC steps). On this page you will run a PRS analysis using PRSice-2, which implements the standard C+T method.

Obtaining PRSice-2

PRSice-2 can be downloaded from:

Operating System	Link
Linux 64-bit	v2.3.3
OS X 64-bit	v2.3.3

and can be directly used after extracting the file.

In this tutorial, you will only need PRSice.R and PRSice_XXX where XXX is the operation system

Required Data

This analysis assumes that you have the following files (or you can download it from [here](#)):

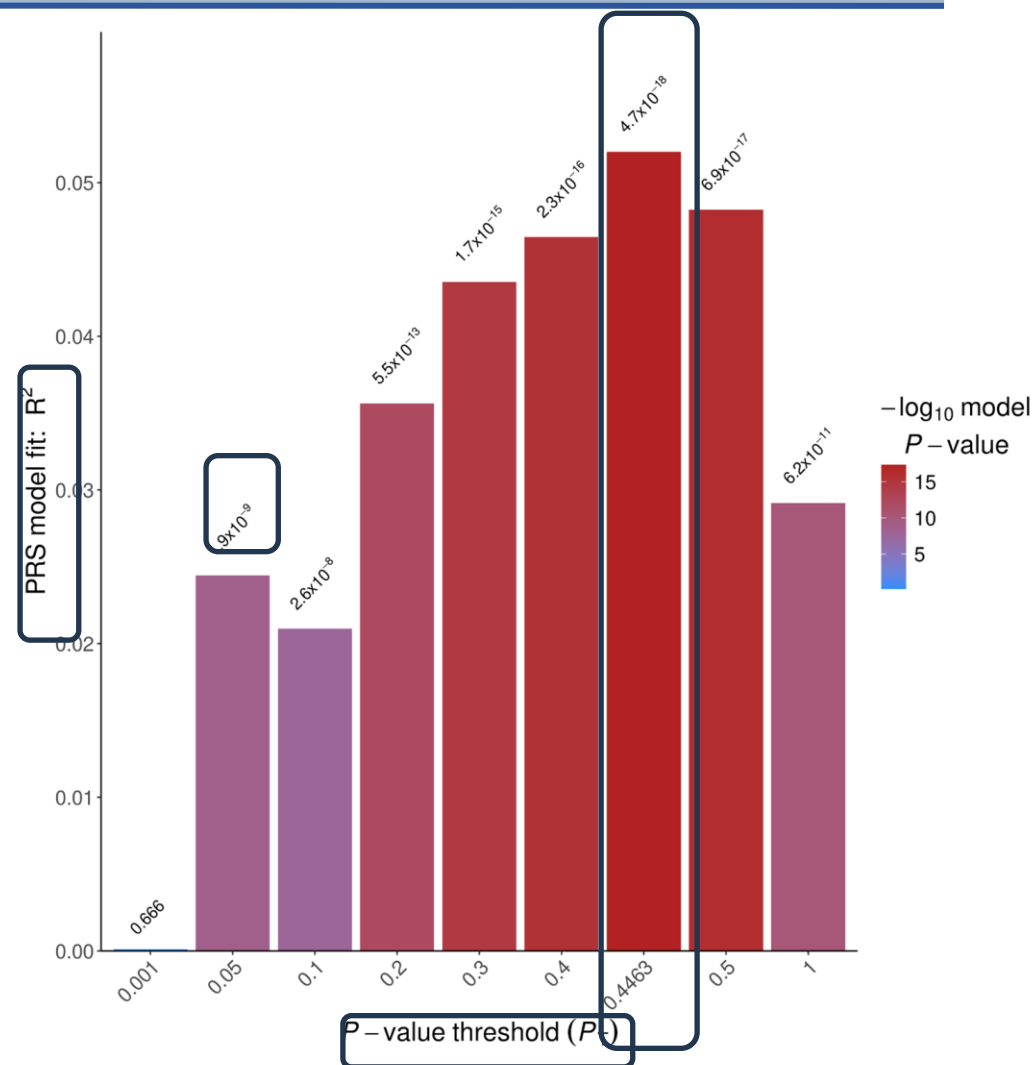
```
wget https://github.com/choishingwan/PRSice/releases/download/2.2.11/PRSice_linux.nightly.zip
unzip PRSice_linux.nightly.zip
```

```
Rscript PRSice.R --dir .
```

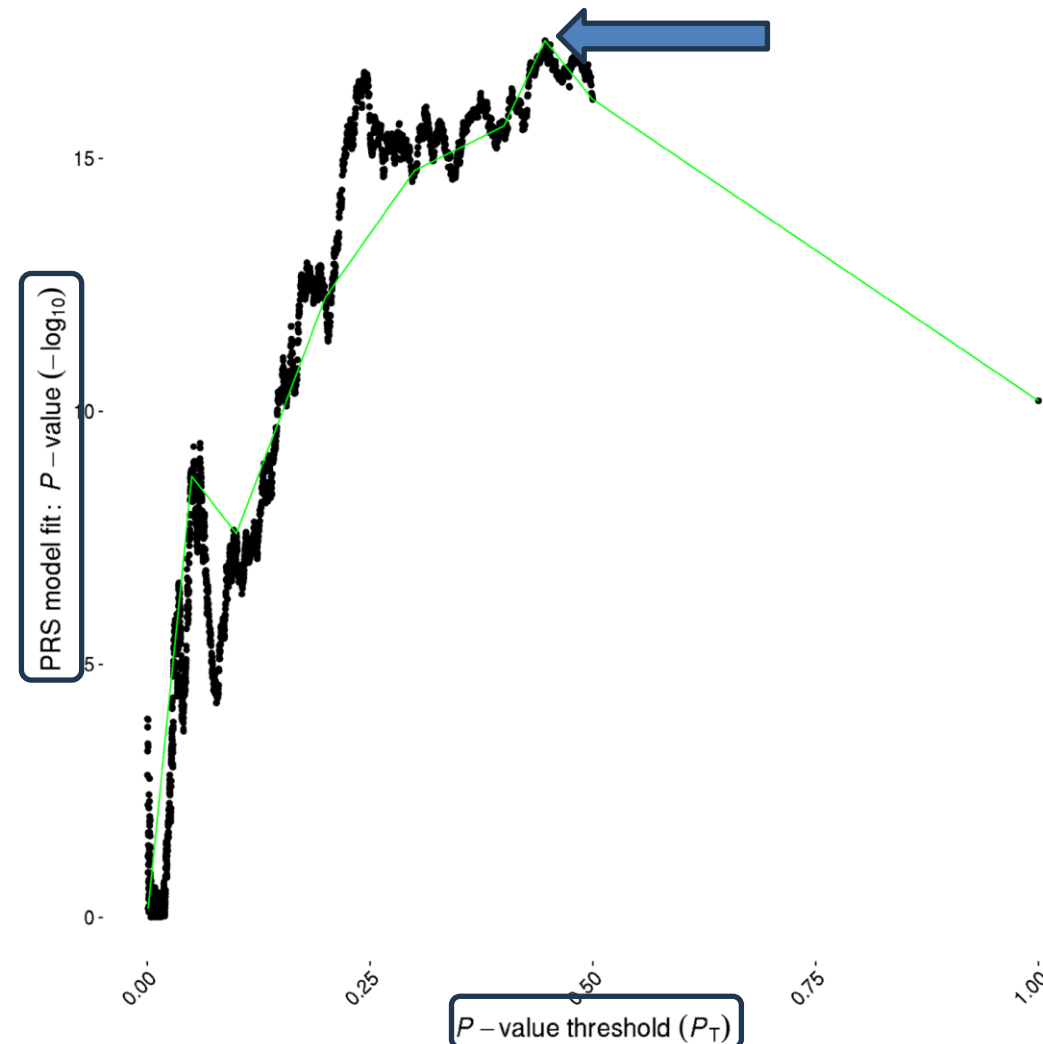
- cd to home directory in terminal
- Navigate to PRSice directory to note the locations of the base and target datasets
- Use the following code to run PRS computation, ensure that you copy the correct locations of the files

```
Rscript /home/as2-streaming-user/PRSice/PRSice.R --dir . \  
--prsice /home/as2-streaming-user/PRSice/PRSice_linux \  
--base /home/as2-streaming-user/PRSice/TOY_BASE_GWAS.assoc \  
--target /home/as2-streaming-user/PRSice/TOY_TARGET_DATA \  
--thread 1 \  
--stat OR \  
--binary-target T
```


- The first plot is PRSice_BARPLOT_<date>.png
- X-axis = p-value threshold for SNP inclusion (pT)
- Y-axis = predictive value, Nagelkerke's R^2
- Each bar shows the model p-value
- Using SNPs with a p-value up to 0.4463 achieves the highest predictive value in the target sample with a p-value of 4.7×10^{-18}



- The second plot is PRSice_HIGH-RES_PLOT_<date>.png
- X-axis = p-value threshold for SNP inclusion (p_T)
- Y-axis = PRS p-values
- The p-value of the predictive effect is in black together with an aggregated trend line in green
- Of note: PRS analysis typically shows that models with lenient p-value thresholds often predict better than models with more stringent thresholds, suggesting that many statistically insignificant SNPs still have predictive value in polygenic traits



```
Rscript /home/as2-streaming-user/PRSize/PRSize.R --dir . \  
--prsice /home/as2-streaming-user/PRSize/PRSize_linux \  
--base /home/as2-streaming-user/data/GCST90027158_buildGRCh38.tsv.gz \  
--target /home/as2-streaming-user/data/ADNI_QC_FINAL \  
--thread 1 \  
--snp variant_id \  
--chr chromosome \  
--bp base_pair_location \  
--A1 effect_allele \  
--A2 other_allele \  
--stat beta \  
--pvalue p_value \  
--bar-levels 5e-8,1e-5,0.1 \  
--binary-target T \  
--fastscore T \  
--out ADNI_PRS
```

- See word document